NOW RECRUITING • GEORGE W. WOODRUFF SCHOOL OF MECHANICAL ENGINEERING

Fully Funded PhD Positions in Experimental Fluid Mechanics

Where fundamental physics meets engineering impact

ABOUT THE GABBARD LAB

The Gabbard Lab is an experimental fluid mechanics and soft matter group in the George W. Woodruff School of Mechanical Engineering at Georgia Tech. Our mission is to uncover the fundamental mechanisms of flowing and soft matter to improve engineering systems. We design, build, and run novel tabletop experiments, paired with physics-focused analysis. Our findings inform industrial heat and mass transfer, guide new environmental remediation techniques, explain natural phenomena, and advance liquid manipulation in capillary-dominated systems.

AREAS OF INTEREST



Formation and transport of complex drops

How complex-fluid drops form from jets and sprays, then spread, rebound, and travel on surfaces, from spray cooling to agriculture.



Dynamic wetting transitions on subfreezing surfaces

How drops spread, stick, or rebound on surfaces below freezing, where wetting races solidification, to guide anti-icing designs.



Particle capture at complex and soft interfaces

How particles are captured, rebound, or escape at drops, films, and soft interfaces, from raindrops scavenging pollutants to wet scrubbers.

WHO SHOULD APPLY

- A B.S. or M.S. in mechanical, aerospace, or chemical engineering, physics, applied math, or a related field
- Curiosity about fluid mechanics, soft matter, and the physics of everyday phenomena
- Enthusiasm for hands-on work: building setups, machining, 3D printing, or electronics
- Some programming for data and image analysis (e.g., Python or MATLAB), or eagerness to learn
- Clear communication, teamwork, and persistence

Prior research experience is helpful, but curiosity and persistence matter most.

WHY GEORGIA TECH

Georgia Tech is a research powerhouse, ranked #2 in the nation for federally sponsored research with $1.43 billion in research expenditures (FY2025). Its campus sits in Midtown Atlanta, a vibrant, diverse city with the world's busiest airport.

BY THE NUMBERS

#1 public graduate mechanical engineering program (#3 overall)

#4 graduate engineering school in the nation

#2 undergraduate mechanical engineering program in the nation

HOW TO APPLY

1. **Explore our research.** Look through our current and past research and publications to make sure the Gabbard Lab aligns with your interests.
2. **Email Chase.** Send your CV, a few sentences about the research you would like to do, and the email addresses of two professional references to chase.gabbard@me.gatech.edu.
3. **Apply to Georgia Tech.** Submit a PhD application to the George W. Woodruff School of Mechanical Engineering by December 15, 2026 and mention the Gabbard Lab in your application. Apply here ↗

AT A GLANCE

START DATE
Fall 2027

FUNDING
Fully funded

APPLY BY
December 15, 2026

CONTACT

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To learn more, visit our website and Google Scholar.

Rankings: U.S. News & World Report, Best Graduate Schools 2026 and Best Colleges 2027. Research expenditures: NSF HERD Survey (2024 data) and Georgia Tech (FY2025).